Turning ride behavior into membership growth

End-to-end customer behavior analysis · 3.82M trips · January-December 2019

BUSINESS QUESTION

How do annual members and casual riders use Cyclistic differently - and where should conversion effort be concentrated?

76.9%
MEMBER SHARE

2.6×
LONGER CASUAL MEDIAN

2.3×
MORE WEEKEND-LED

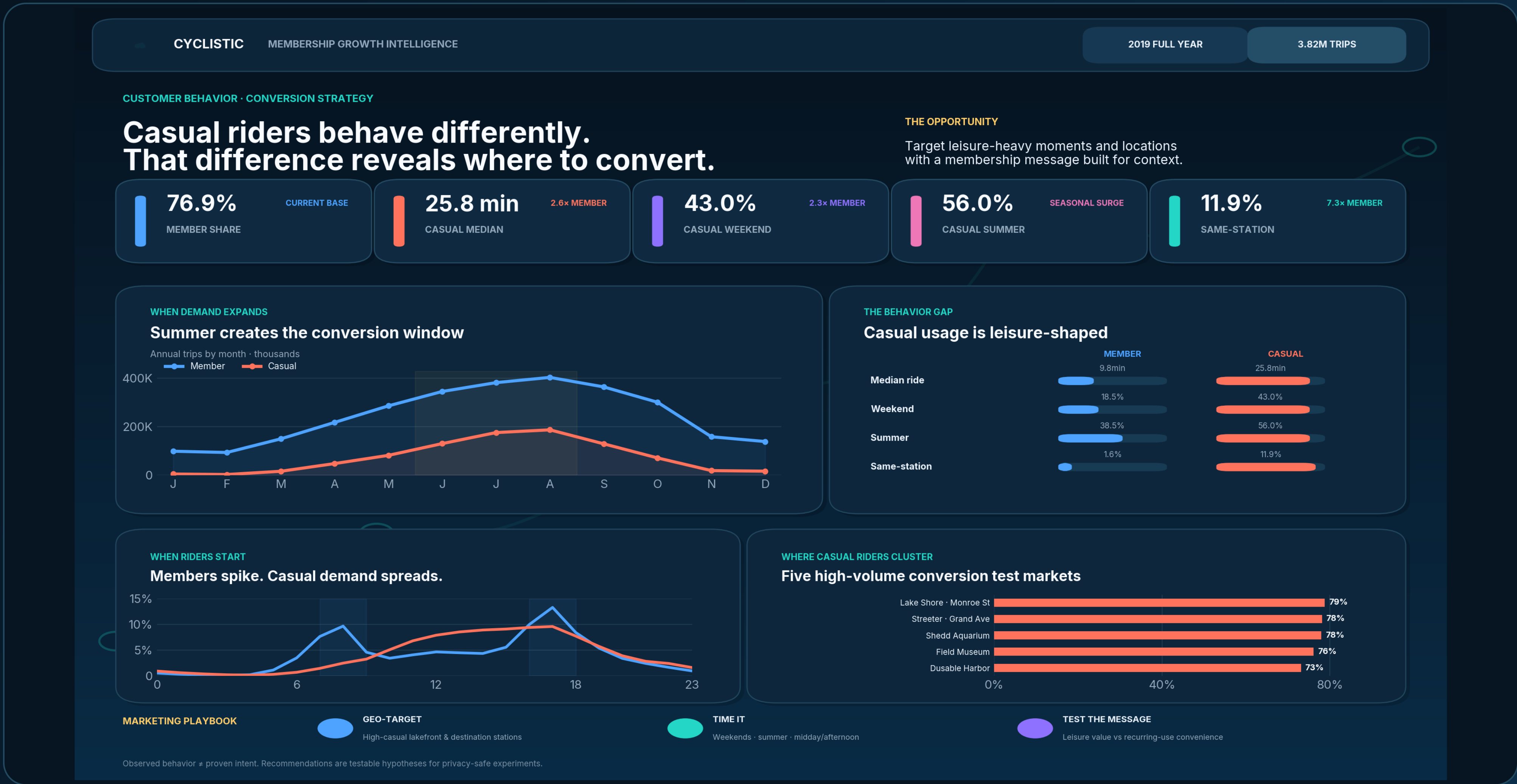
7.3×
MORE SAME-STATION

Python · Pandas · NumPy · SQL · data quality · behavioral analytics · executive storytelling

THE DECISION IN ONE PAGE

Casual riders behave like a different use case - so the marketing should too.

Across 3.82M rides, casual users take longer trips, concentrate more heavily on weekends and summer, and return to the same station far more often. Those patterns point to context-specific acquisition rather than a generic membership message.



01 · FRAME THE PROBLEM

Find the behavioral differences that can guide membership conversion.

BUSINESS TASK

How do members and casual riders use Cyclistic differently?

Translate measurable differences in when, where and how long people ride into marketing opportunities that can be tested.

STAKEHOLDERS

Marketing director + executive team

The output needs to balance analytical rigor with concise, decision-ready communication.

SUCCESS CRITERIA

Evidence → implication → action

Every recommendation must connect directly to observed trip behavior and remain explicit about what the data cannot prove.

WHY 2019?

The most recent complete 12-month window in the supplied archive.

The uploaded archive spans multiple years but has gaps in later monthly files. Using January-December 2019 avoids missing-month seasonality bias and preserves a complete annual cycle.

Source	Rows	Role
Q1	365,069	Winter / early spring
Q2	1,108,163	Spring / early summer
Q3	1,640,718	Peak summer
Q4	704,054	Fall / winter

DATA DISCIPLINE

3.82M

trip-level records harmonized across four quarterly files.

Privacy by design

Gender and birth-year fields were not used because they are unnecessary for the business question. The analysis is behavioral and aggregate.

02 · BUILD TRUST IN THE DATA

The pipeline preserves valid rides instead of blindly deleting anomalies.

1

Ingest

Load four quarterly files and preserve raw row counts.

→

2

Harmonize

Map Q2 schema differences and standardize rider types.

→

3

Validate

Check IDs, timestamps, stations, categories and duration logic.

→

4

Engineer

Create duration, month, weekday, hour, season and route features.

→

5

Audit

Export transparent quality metrics and sensitivity summaries.

QUALITY AUDIT

Check	Count	Treatment
Duplicate ride IDs	0	No action needed
Invalid timestamps	0	No action needed
Missing station names	0	No action needed
Non-positive calculated durations	13	Recovered from source duration
Rides >24 hours	1,849	Retained for volume; flagged for duration analysis

13 rides looked impossible. They were not.

On 3 November 2019, the daylight-saving fallback makes local clock subtraction produce negative durations for 13 records. The source `tripduration` remains positive, so those rides are recovered rather than discarded.

Why median duration is the headline KPI

1,849 rides exceed 24 hours. They are legitimate volume observations but can heavily distort the mean, so median ride duration is the primary comparison and capped-duration sensitivity is retained separately.

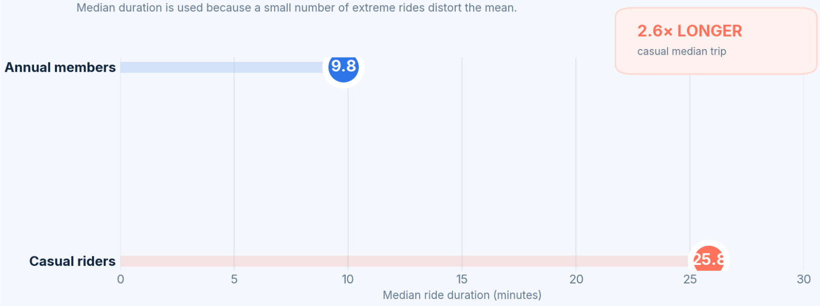
03 · TWO SIGNALS IMMEDIATELY SEPARATE THE SEGMENTS

Casual riders stay out longer and ride far more often on weekends.

BEHAVIOR GAP

Casual trips are not just longer - they are 2.6× longer

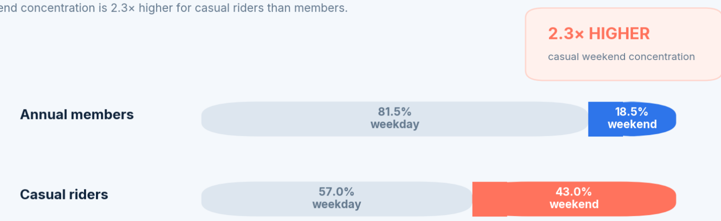
Median duration is used because a small number of extreme rides distort the mean.



USAGE CONTEXT

Casual demand shifts decisively toward the weekend

Weekend concentration is 2.3× higher for casual riders than members.



2.6× longer median ride

Casual median duration is 25.8 minutes versus 9.8 minutes for members. The gap persists despite outlier controls.

2.3× higher weekend concentration

43.0% of casual trips occur on weekends versus 18.5% for members. Saturday is the casual segment's peak weekday.

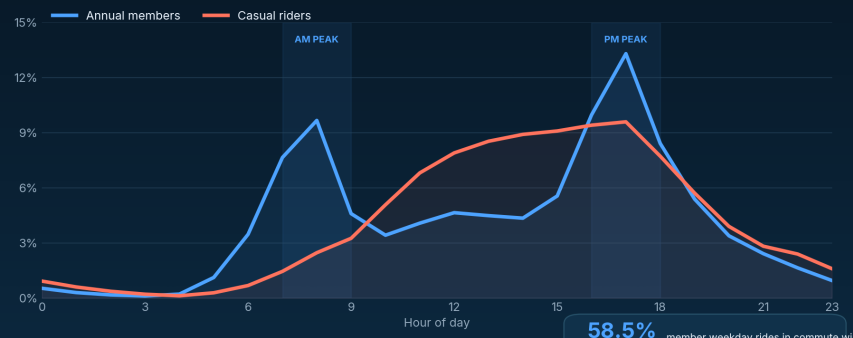
04 · DEMAND TIMING

Member use looks recurring. Casual use looks seasonal and experience-led.

TIME-OF-DAY SIGNAL

Members cluster around commute windows

Casual riders build gradually through midday; member demand forms sharp morning and evening peaks.

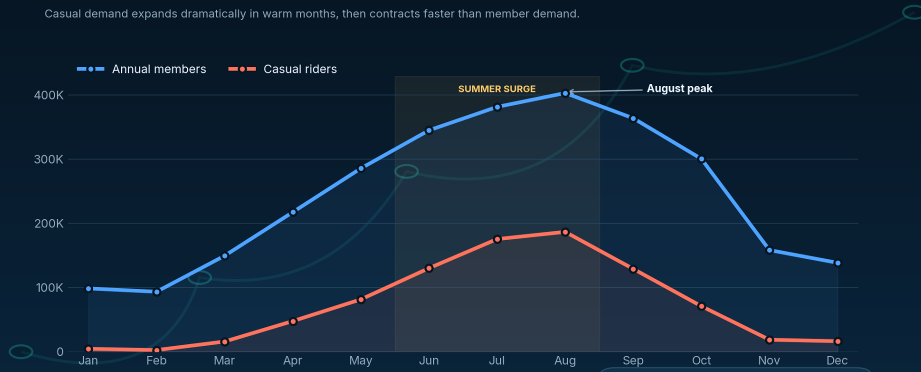


58.5% member weekday rides in commute windows

SEASONALITY

Summer is the growth window

Casual demand expands dramatically in warm months, then contracts faster than member demand.



56.0% of casual rides occur Jun-Aug

58.5% of member weekday rides

start during the defined 7-9am and 4-6pm windows, versus 37.4% for casual riders. This is consistent with recurring weekday transport - not proof of trip purpose.

56.0% of casual rides

occur in June-August, compared with 38.5% of member rides. Summer is the most obvious acquisition window.

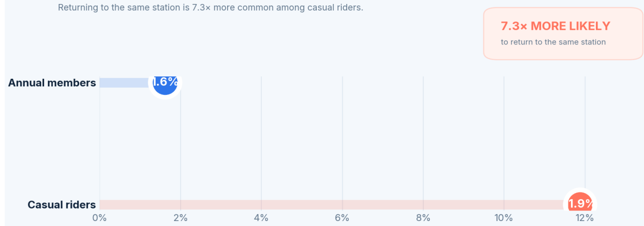
05 · GEOGRAPHY AND ROUTE SHAPE

Round-trip behavior and destination hotspots reinforce the leisure-use pattern.

ROUTE BEHAVIOR

Round-trip behavior is a defining casual signal

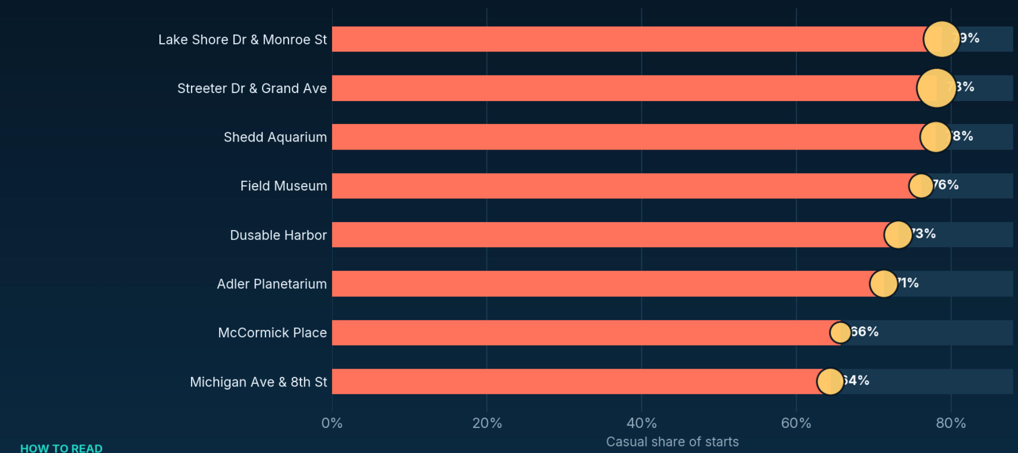
Returning to the same station is 7.3× more common among casual riders.



GEO OPPORTUNITY

Casual demand clusters around destination stations

High-volume stations with the strongest casual share create natural test markets for conversion campaigns.



HOW TO READ

Bar = casual share · Gold bubble = relative annual station volume

Top signal: Lake Shore Dr & Monroe St · 79% casual

7.3× higher same-station rate

11.9% of casual rides return to the starting station, compared with 1.6% of member rides.

High-casual test markets

Lake Shore Dr & Monroe St, Streeter Dr & Grand Ave, Shedd Aquarium, Field Museum and Dusable Harbor each show casual shares above 70% in this dataset.

06 · RECOMMENDATIONS

Three moves - each tied to an observed behavior and a measurable test.

01

Geo-target high-casual stations

Prioritize lakefront, museum and destination stations where casual demand is already concentrated. Test QR prompts, in-app messaging and localized paid social.

PRIMARY KPI · conversion rate

02

Time campaigns to casual demand

Concentrate acquisition around weekends, summer, and midday/afternoon periods instead of mirroring member commute patterns.

PRIMARY KPI · incremental member starts

03

Segment the value proposition

Test leisure-oriented membership value against convenience/regular-use messaging. Match creative to the behavioral context.

PRIMARY KPI · lift by segment

RECOMMENDED TEST DESIGN

Segment / context	Treatment hypothesis	Measurement
Weekend · high-casual station	Leisure value + conversion incentive	Membership conversion rate
Weekday · commute-time casual	Recurring-use convenience + membership economics	Incremental conversion lift
Control	Business-as-usual / generic message	Baseline conversion

ANALYTICAL MATURITY

Strong descriptive evidence. No invented causality.

Limitations

Historical window: 2019 behavior should be refreshed on a complete recent 12-month window before operational decisions.

Trip-level data: repeat casual riders and eventual membership conversions cannot be identified.

Missing causal drivers: no campaign exposure, weather, events, pricing sensitivity or explicit trip-purpose fields are available.

Interpretation: timing and station patterns support targeting hypotheses, but they do not prove intent.

NEXT ANALYTICAL ROADMAP

Move from behavioral segmentation to incremental lift.

1. Refresh with a complete recent year.
2. Connect privacy-safe campaign exposure and conversion outcomes.
3. Run controlled A/B tests by station/time segment.
4. Estimate incremental members and cost per incremental member.

Portfolio signal

The value of this project is not the number of charts. It is the reproducible workflow, explicit assumptions, data-quality judgment, and ability to translate analysis into a testable business decision.

Python
PANDAS · NUMPY
Harmonization + feature engineering

SQL
BUSINESS QUERIES
Aggregations + validation

BI
EXECUTIVE STORYTELLING
Dashboard-ready outputs

GitHub
REPRODUCIBILITY
Notebook + docs + code